

1 Highlights

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- 5 • Robust spatiotemporal framework for accurate water-quality reconstruction.
- 6 • Auxiliary-variable node enhancement for better spatial adaptability.
- 7 • Outperformed state-of-the-art baselines in accuracy and robustness.

Water Quality Spatial-Temporal Imputation Using Diffusion Graph Convolutional Networks: A Case Study in Georgia, USA

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ARTICLE INFO

Keywords:

Water quality

pH imputation

Spatio-temporal kriging

Graph Convolutional Networks

ABSTRACT

Water quality monitoring is essential for environmental management and public health, yet the sparsity of sensor deployments often leads to missing spatio-temporal data, necessitating accurate imputation techniques to infer water quality at unobserved monitoring locations. Conventional methods rarely capture water quality's spatio-temporal, multi-parameter dynamics or handle noisy, evolving sensor networks. To address this, a novel hybrid framework—the Water Quality Auxiliary-Enhanced Spatio-Temporal Imputation Framework based on Diffusion Graph Convolutional Networks is proposed. This framework employs diffusion graph convolutional networks to model the spatio-temporal correlations of pH data and adaptively integrates auxiliary variables to generate node-specific enhancement signals. In addition, a subgraph-sampling design enables to address the influence of noisy neighboring nodes and facilitate generalization across dynamic monitoring networks. Extensive experiments on a real-world daily water-quality dataset from 37 monitoring locations in Georgia, USA demonstrate that our model outperforms state-of-the-art baselines; on the test set, it achieves an MAE of 0.0084, MAPE of 1.27%, and RMSE of 0.0119 for pH imputation, reducing these metrics by 56.0%, 57.0%, and 51.6% relative to the best baseline. These results suggest that the proposed framework provides a practical and scalable tool for spatio-temporal water-quality imputation at unobserved monitoring locations in river networks. The experimental data and codes for this study can be found at <https://github.com/Xielewei/Water-Quality.git>.

1. Introduction

With the continuous acceleration of industrialization and urbanization, water pollution has become an increasingly severe environmental problem, posing a significant threat to ecosystem stability and public health (Xie, Zhu, Xie, Bu and Zhou (2025); Saadat, Emami, Hamidifar and Shilsar (2025); Tan, Bai, Yue and Jia (2025)). As a core component of environmental monitoring and water resource management, water quality assessment relies on the accurate and continuous acquisition of monitoring data (Pati, Tripathy and Subudhi (2025); Raheli, Talabbeydokhti and Nourani (2024)). Accurate analysis of key water quality parameters (e.g., pH, dissolved oxygen, turbidity, and conductivity) enables scientific support for pollution tracing, environmental assessment, and management decision-making (Zahran, Zhang, Tango and Smith (2022); Huang, Cai, He, Dai, Wan and Guo (2024)).

However, in real-world monitoring scenarios, water quality data often exhibit spatio-temporal missingness due to sparse sensor deployment, high maintenance costs, and susceptibility of monitoring equipment to environmental and operational disturbances (Banjara, Poudel, Pariyar, Upreti, Zafeirakou and Paudel (2025); Zhang, Wang, Liu, Wei, Zhang, Huang and Liu (2024)). Such missing data not only reduce the reliability of monitoring results but also significantly affect subsequent tasks such as water quality prediction, pollution detection, and early warning analysis (Wu, Shan, Recknagel, Wang and Shang (2025)). As shown in Figure 1, the monitoring stations are unevenly distributed

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53 along the river network, with several stations suffering from missing or incomplete data records due to environmental
 54 and operational factors.

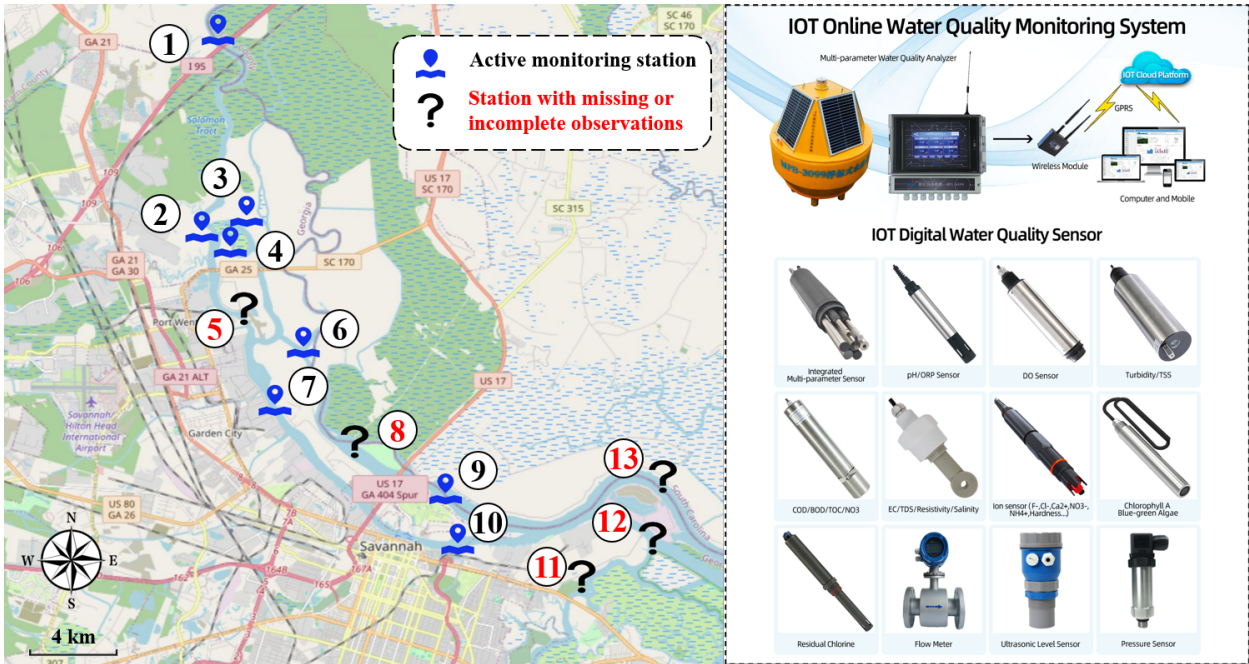


Figure 1: Spatial distribution of water quality monitoring stations and schematic of the IoT-based online monitoring system. (The left panel shows 13 monitoring stations along the river, while the right panel depicts the multi-parameter IoT system for real-time data acquisition. Blue labels indicate active stations, and red labels with question marks denote missing or incomplete observations.)

55 To mitigate these issues, spatio-temporal data imputation has emerged as an essential research direction in
 56 intelligent water environment monitoring. High-precision reconstruction of missing data can effectively restore the
 57 continuity and completeness of monitoring datasets, thereby providing a reliable foundation for subsequent analysis,
 58 modeling, and decision support (Velayudhan, Devidas and Savić (2025)). Traditional imputation methods, such as
 59 statistical interpolation and Kriging, perform well under regular spatial structures but struggle to capture complex
 60 nonlinear dependencies and multi-scale spatio-temporal correlations (Wang, Li and Chen (2025); Huan (2023)).

61 In recent years, deep learning techniques—particularly Graph Neural Networks (GNNs)—have demonstrated
 62 superior performance in handling irregular spatial structures. In particular, Khorsandi, Mahdavi, Salajeghe and
 63 Eslamian (2011) showed that the neural network outperformed other traditional methods in reconstructing monthly
 64 precipitation data. Several studies like Xu, Zhang, Zhang, Shao, Yu, Chu and Qian (2025); Yuan and Lin (2025)
 65 have modeled spatial topological relationships and temporal dependencies among monitoring nodes through graph-
 66 based structures, achieving accurate predictions in complex environments. Meanwhile, Cini, Marisca and Alippi (2021)
 67 introduced the GRIN framework that learns spatio-temporal representations via message passing and achieves over 20%
 68 improvement in mean absolute error compared to state-of-the-art methods; Gordon, Petousis, Zheng, Zamanzadeh
 69 and Bui (2021) incorporates temporal information into bipartite graphs to effectively recover missing data under
 70 different missing rates; Chen, Wang, Lei, He, Fu and Lu (2022) proposes an adaptive graph learning layer that jointly
 71 captures spatial and temporal dependencies without prior topology, demonstrating superior reconstruction accuracy in
 72 environmental sensor networks.

73 Additionally, spatio-temporal imputation methods have been extensively explored in other environmental domains.
 74 For example, temporal-imputation-based frameworks have been proposed for wave height forecasting (Si, Wang and
 75 Deng (2025)), while Transformer architectures have been employed to perform predictions directly under missing
 76 data conditions (Nejad, Alaiz-Rodríguez, McCarthy, Kelleher, Grey and Parnell (2024)). Similarly, graph convolution
 77 mechanisms have been utilized to reconstruct spatial interaction relationships, achieving strong performance in urban
 78 traffic and environmental data restoration (Xiao, Gong, Wang, Di and Jing (2024)). Meanwhile, Betancourt, Li, Kleinert

and Schultz (2023) applied the “Correct and Smooth” algorithm for hybrid graph learning, significantly reducing imputation error in ozone monitoring data from 278 stations across Germany; Chen, Yang, Yang, Peng and Ge (2025) proposed an adaptive spatio-temporal feature fusion layer that enhances robustness under varying missingness ratios; You, Ma, Ding, Kochenderfer and Leskovec (2020) modeled both observations and features as nodes in a bipartite graph to jointly perform imputation and downstream prediction.

Despite the significant progress achieved by graph-based and deep spatio-temporal imputation methods, existing studies in the context of water quality monitoring still exhibit several limitations.

1. **Lack of dedicated modeling for the unique characteristics of water quality data.** Most existing studies have been developed primarily for air quality or traffic-related datasets, whereas water quality measurements typically exhibit stronger nonlinearity, multi-source heterogeneity, hydrological dependencies, and pronounced seasonal variations. Moreover, key parameters exhibit complex internal correlations (Zhu, Qiao, Li, Zhang, Han, Shen, Yuan and Chen (2026)), yet current methods generally fail to fully exploit these multi-parameter relationships or incorporate water-quality-specific mechanisms.
2. **Single-dimensional imputation that lacks joint spatio-temporal variable inference.** Many existing frameworks rely either on temporal interpolation using individual sensor time series or on spatial inference based on graph neighborhoods. However, they pay limited attention to collaborative imputation across multiple time periods, multiple monitoring stations, and multiple water quality parameters, which restricts their applicability in real-world water quality monitoring (Lin, Lim and Oh (2024)).
3. **High sensitivity to neighborhood reliability and graph structural stability.** Graph-based imputation methods typically assume that neighboring nodes provide stable and complete observations. In practice, monitoring locations frequently suffer from sensor malfunction, drift, or communication interruptions, resulting in noisy or missing values (Xia, Zhang, Lansey and Liu (2025)). Such unreliability can disrupt the information propagation process and introduce instability into model (Santos-Fernandez, Denman and Mengersen (2025)). Furthermore, monitoring networks may undergo temporary sensor additions or removals during operation, and models relying on fixed graph structures generally require graph reconstruction and retraining to accommodate such changes, thereby limiting their adaptability.

Building upon the above limitations, achieving high-quality spatio-temporal imputation for real-world water quality monitoring data remains challenging, including accurately capturing cross-station spatial dependencies in irregular river network structures, enabling collaborative modeling across multiple temporal scales and water quality parameters, and maintaining stable and robust information propagation in the presence of neighborhood noise, missing observations, and dynamic changes in the monitoring network topology.

To address these challenging, this paper proposes a water quality auxiliary-enhanced spatio-temporal imputation model based on a Diffusion Graph Convolutional Network (DGCN) (Zheng, Gao, Chen, Jin and Li (2021)). The proposed method precisely models spatio-temporal dependencies using diffusion graph convolution, while incorporating multiple auxiliary water quality indicators (e.g., dissolved oxygen, turbidity, conductivity) as node-specific enhancement signals to adaptively strengthen feature representation across nodes. In addition, an inductive subgraph sampling strategy is adopted to mitigate the impact of noisy or unreliable neighboring nodes and to enable generalization to dynamically changing monitoring networks. To provide a more intuitive illustration of the technical pipeline of this study, Fig. 2 presents the overall research flowchart, including the task scenario, dataset, experimental procedure, and result analysis.

The main contributions of this study are summarized as follows:

1. A robust spatio-temporal water quality imputation framework is proposed, capable of effectively reconstructing key water quality parameters under missing data scenarios;
2. A node-level enhancement mechanism based on auxiliary variables is introduced, enabling diffusion graph convolution to exploit joint spatio-temporal dependencies and cross-variable correlations for improved spatial adaptability and generalization;
3. A subgraph sampling strategy is incorporated to reduce sensitivity to unreliable neighborhoods and structural instability, enabling more robust inference in sparsely monitored sensor networks;
4. Extensive experiments on real-world water quality datasets demonstrate that the proposed method outperforms existing state-of-the-art models in terms of imputation accuracy and robustness.

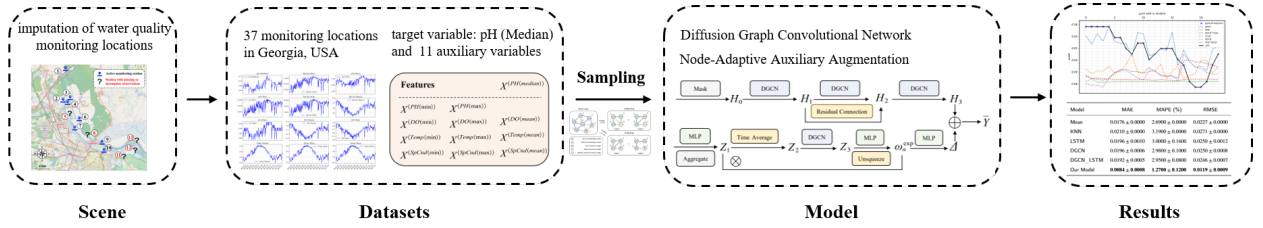


Figure 2: Overall research flowchart.

Table 1

Summary of main notations at the global and sample levels.

Symbol	Description
Global level	
F	Number of water quality indicators
H	Length of the global time horizon
N	Number of observed monitoring locations
U	Number of unobserved monitoring locations
Sample / Subgraph level	
h	Length of the time window for each sample
n	Number of observed nodes in a sampled subgraph
u	Number of unobserved nodes in a sampled subgraph
N_{sub}	Number of nodes in a sampled subgraph ($n + u = N_{\text{sub}}$)

2. Methods

2.1. Problem definition

In a spatio-temporal water quality monitoring system, we collect F water quality indicators at N observed monitoring locations over a global time horizon of length H . The complete dataset at the observed locations is denoted by $\mathcal{X} = \{X^{(f)} \in \mathbb{R}^{N \times H} \mid f = 1, 2, \dots, F\}$, where each matrix $X^{(f)}$ describes the spatial pattern and temporal variation of feature f across all observed sites and time steps. In this study, we designate the median power of hydrogen (pH) value as the imputation target, represented by the matrix $X \in \mathbb{R}^{N \times H}$ (corresponding to $X = X^{(1)}$), and treat the remaining $F - 1$ indicators as auxiliary variables to form the auxiliary dataset $\mathcal{X}_{\text{aux}} = \{X^{(f)} \in \mathbb{R}^{N \times H} \mid f = 2, 3, \dots, F\}$, which provides additional information to support the imputation.

Beyond the N observed monitoring locations equipped with sensors, the monitoring network may contain U unobserved monitoring locations without direct pH measurements. Spatial connectivity among all $N + U$ locations is represented by an adjacency matrix $A \in [0, 1]^{(N+U) \times (N+U)}$, which is constructed using a thresholded Gaussian kernel based on the geographic distances between locations. Our overall goal is to generate pH time series for the U unobserved monitoring locations, yielding a predicted matrix $\hat{Y} \in \mathbb{R}^{U \times H}$ that approximates the unknown true matrix $Y \in \mathbb{R}^{U \times H}$. Formally, we aim to learn a mapping f_{θ} such that

$$\hat{Y} = f_{\theta}(X, \mathcal{X}_{\text{aux}}, A) \approx Y,$$

given the observed spatio-temporal signals at the N sensor locations and the graph structure over all $N + U$ locations.

To clarify the dimensions and notations introduced above, we summarize the main symbols related to features, time horizons, and monitoring locations at both the global and sample levels in Table 1.

2.2. Subgraph Sampling and Masking

In the spatio-temporal water quality imputation task, we introduce a subgraph sampling and masking mechanism that trains the model to learn spatial message-passing strategies, thereby enabling inductive imputation. Compared with traditional transductive methods (e.g., matrix or tensor completion) that are tied to a fixed set of sensors, this design

151 naturally accommodates dynamic monitoring networks in which sensors may be added, retired, or relocated. During
 152 inference, the construction of subgraphs can further prioritize reliable observed monitoring locations as context nodes,
 153 which helps mitigate the influence of noisy measurements while still exploiting the underlying graph structure.

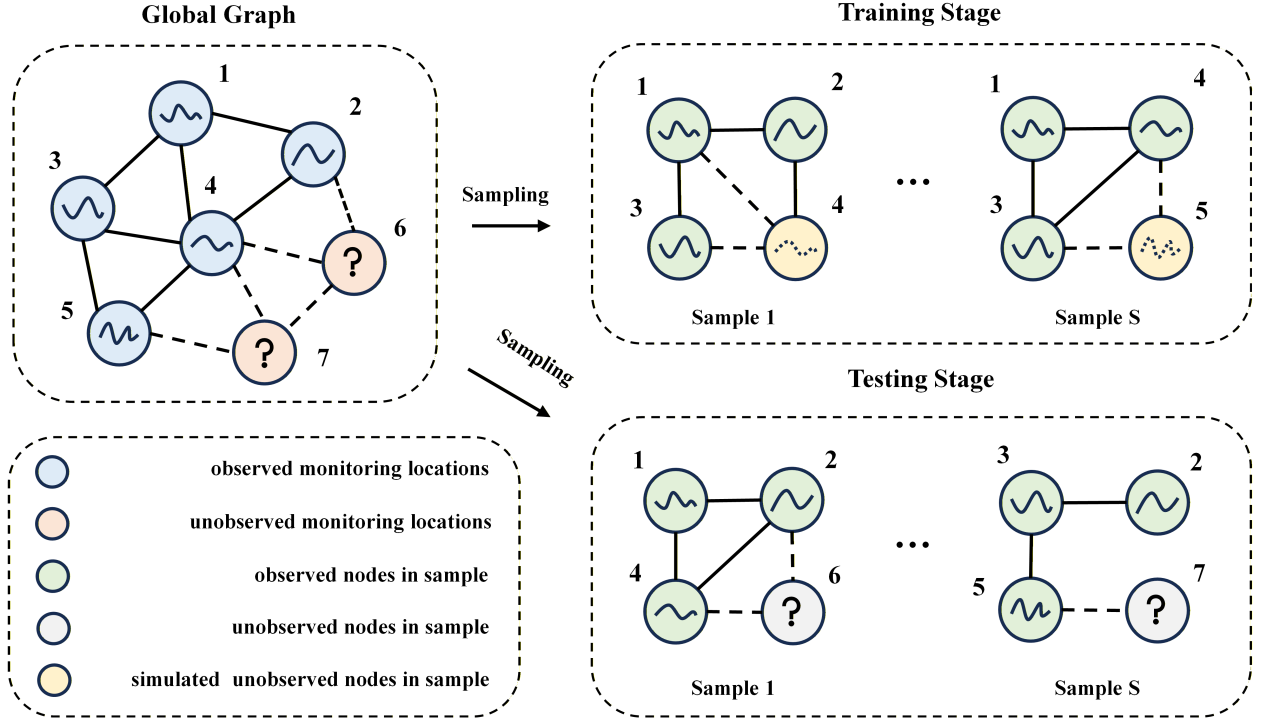


Figure 3: Subgraph Sampling

154 Figure 3 illustrates the subgraph sampling strategy for the global water quality monitoring network. The left
 155 panel shows the global graph composed of N observed monitoring locations and U unobserved monitoring locations.
 156 Observed monitoring locations (nodes 1–5 in the example) are equipped with sensors and provide historical pH and
 157 auxiliary measurements, whereas unobserved monitoring locations (nodes 6 and 7) do not have pH observations. To
 158 encode the availability of pH measurements over time, we define a binary mask matrix $M \in \{0, 1\}^{(N+U) \times H}$, where
 159 $M_{i,t} = 1$ indicates that the pH value at location i and time t is observed and $M_{i,t} = 0$ otherwise. For unobserved
 160 monitoring locations such as nodes 6 and 7, the corresponding rows of M are zero at all time steps because pH is never
 161 measured at these locations.

162 During the *training stage*, subgraphs are sampled only from the set of observed monitoring locations. At each
 163 sampling step, we randomly select a subset of $n + u$ nodes from $\{1, \dots, N\}$ to form a node index set $I_{\text{sub}} =$
 164 $\{i_1, \dots, i_n, i_{n+1}, \dots, i_{n+u}\}$. The first n nodes in I_{sub} are designated as observed nodes in the current subgraph, while
 165 the remaining u nodes are treated as *simulated unobserved nodes*: they correspond to observed monitoring locations
 166 in the original graph, but their pH time series are deliberately masked to construct supervised missing-data scenarios.
 167 In Training Sample 1 of Figure 3, for example, nodes $\{1, 2, 3, 4\}$ form a training subgraph, where nodes 1–3 act as
 168 observed nodes and node 4 is used as a simulated unobserved node.

169 A temporal window of length h is then drawn by choosing a starting index j and defining $J_{\text{sub}} = [j, j + h - 1]$.
 170 The corresponding pH values, auxiliary features, and adjacency entries are extracted to form the subgraph inputs:

$$X_{\text{sub}}^{(s)} = X[I_{\text{sub}}, J_{\text{sub}}], \quad \mathcal{X}_{\text{aux,sub}}^{(s)} = \{X^{(f)}[I_{\text{sub}}, J_{\text{sub}}]\}_{f=2}^F, \quad A_{\text{sub}}^{(s)} = A[I_{\text{sub}}, I_{\text{sub}}].$$

The subgraph mask $M_{\text{sub}}^{(s)} \in \{0, 1\}^{(n+u) \times h}$ is obtained by slicing the global mask M and then overwriting the rows associated with simulated unobserved nodes:

$$M_{\text{sub}}^{(s)}[1 : n, :] = 1, \quad M_{\text{sub}}^{(s)}[n + 1 : n + u, :] = 0.$$

In addition, the pH entries of simulated unobserved nodes in $X_{\text{sub}}^{(s)}$ are set to zero so that the model must infer them from the remaining observed nodes and the local graph structure. Repeating this procedure over many random node combinations and time windows yields a collection of training samples

$$S_{\text{train}} = \{(X_{\text{sub}}^{(s)}, \mathcal{X}_{\text{aux,sub}}^{(s)}, M_{\text{sub}}^{(s)}, A_{\text{sub}}^{(s)}) \mid s = 1, \dots, S\},$$

on which the model is optimized.

During the *testing (inference) stage*, subgraphs are constructed on the full set of $N + U$ nodes. Each test subgraph contains one or more true unobserved nodes together with a set of observed nodes that provide contextual information. In Test Sample 1 of Figure 3, for instance, nodes $\{1, 2, 4, 6\}$ form a test subgraph sample, where nodes 1, 2, and 4 are observed nodes and node 6 is an unobserved node; another test sample may use nodes $\{2, 3, 5, 7\}$, where node 7 is an unobserved node. For every unobserved node in a test subgraph, its pH entries in the input $X_{\text{sub}}^{(s)}$ are set to zero and the corresponding rows in $M_{\text{sub}}^{(s)}$ are set to 0, while its auxiliary features and graph connectivity are preserved.

To be noted, the node subset I_{sub} is sampled uniformly at random in each iteration during training, exposing the model to diverse node combinations and improving its ability to generalize to unseen graph configurations. At test time, one may continue to select observed nodes uniformly at random around each target unobserved node, or alternatively prioritize nearby and more reliable observed monitoring locations as context nodes, thereby reducing the impact of noisy measurements while still leveraging the spatial structure. The subgraph size parameters n and u can be chosen flexibly: n is typically set larger than u so that each subgraph contains a sufficient number of observed nodes, but the method does not rely on a fixed ratio and allows n and u to be adapted to the density of the monitoring network.

2.3. Model Architecture

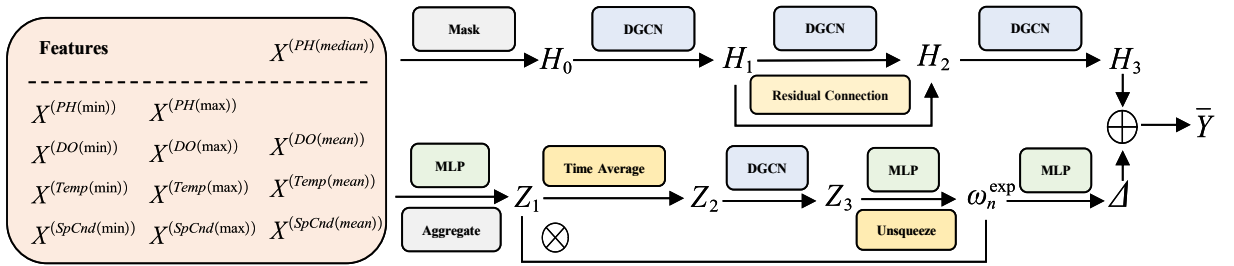


Figure 4: Model Framework

Given the remarkable performance of IGNNK (Wu, Zhuang, Labbe and Sun (2021)) in spatiotemporal kriging tasks, we design a three-layer Diffusion Graph Convolutional Network (DGCN) as the core architecture, integrated with a node-adaptive augmentation module to effectively fuse auxiliary features, showing in Figure 4. The model first learns spatial message passing mechanisms from the masked pH subgraph signals $X_{\text{sub}}^{(s)} \odot M_{\text{sub}}^{(s)}$ via DGCN, yielding preliminary reconstructions $\hat{X}_{\text{sub}}^{(s)}$; subsequently, it generates node-specific augmentation signals $\Delta^{(s)}$ using the auxiliary subset $\mathcal{X}_{\text{aux,sub}}^{(s)}$, producing enhanced reconstructions $\tilde{X}_{\text{sub}}^{(s)} = \hat{X}_{\text{sub}}^{(s)} + \Delta^{(s)}$. This layered design captures asymmetric spatial dependencies driven by geographic distances while adaptively elevating the contribution of auxiliary variables to pH imputation. The architecture is parameter-efficient, supports batch processing of subgraph samples S , and optimizes overall reconstruction error during training.

2.4. Diffusion Graph Convolutional Network

The DGCN serves as the foundational building block, employing diffusion convolution operations to handle the directionality and geographic distance effects in water quality monitoring networks Li, Yu, Shahabi and Liu (2017). Given the subgraph adjacency matrix $A_{\text{sub}}^{(s)} \in [0, 1]^{(n+u) \times (n+u)}$, we first compute the forward random walk matrix

$$\bar{A}_f^{(s)} = D_f^{-1} A_{\text{sub}}^{(s)}, \quad \bar{A}_b^{(s)} = D_b^{-1} (A_{\text{sub}}^{(s)})^\top \quad (1)$$

where D_f and D_b are row and column wise degree normalization matrices. This formulation ensures that message passing captures “reachability” from observed to simulated unobserved nodes.

The core diffusion process of DGCN is computed via K-order Chebyshev polynomial approximation:

$$H_{l+1} = \sum_{k=1}^K \left(T_k(\bar{A}_f^{(s)}) H_l \Theta_{k,f,l} + T_k(\bar{A}_b^{(s)}) H_l \Theta_{k,b,l} \right) \quad (2)$$

where $T_k(\cdot)$ is recursively defined as $T_0(X) = I$, $T_1(X) = X$, $T_k(X) = 2XT_{k-1}(X) - T_{k-2}(X)$; and $\Theta_{k,f,l}$, $\Theta_{k,b,l}$ are learnable parameter matrices controlling inter-node information transformations. The diffusion order K balances computational efficiency with spatial dependency capture.

The model architecture comprises three stacked DGCN layers, with the first layer projecting the input $H_0 = X_{\text{sub}}^{(s)} \odot M_{\text{sub}}^{(s)} \in \mathbb{R}^{(n+u) \times h}$ to the hidden space via

$$H_1 = \text{DGCN}(H_0, \bar{A}_f^{(s)}, \bar{A}_b^{(s)}), \quad (3)$$

$$H_2 = \sigma \left(\text{DGCN}(H_1, \bar{A}_f^{(s)}, \bar{A}_b^{(s)}) \right) + H_1, \quad (4)$$

$$H_3 = \text{DGCN}(H_2, \bar{A}_f^{(s)}, \bar{A}_b^{(s)}), \quad (5)$$

where σ denotes the ReLU activation. The dimensions of the hidden representations are as follows: $H_1, H_2 \in \mathbb{R}^{(n+u) \times d_h}$, where d_h is the hidden dimension, and $\hat{Y}_{\text{sub}}^{(s)} = H_3 \in \mathbb{R}^{(n+u) \times h}$, matching the input feature dimension h . The residual connection in the second layer adds the output of the activated DGCN transformation back to the input H_1 , preserving raw node information and thereby enhancing generalization in reconstructing missing pH values.

2.5. Node-Adaptive Auxiliary Augmentation

To fully leverage the auxiliary features, we introduce a node-adaptive augmentation module. This module encodes them into node-specific augmentation signals $\Delta^{(s)}$. It generates node preference weights via linear transformations and temporal aggregation, preventing auxiliary variables from dominating the pH reconstruction.

The auxiliary subset $\mathcal{X}_{\text{aux,sub}}^{(s)} \in \mathbb{R}^{(n+u) \times h \times (F-1)}$ is first encoded through a linear transformation followed by a ReLU activation, then aggregated over the temporal dimension to extract node-level representations, and finally enhanced via spatial propagation on the graph structure. Specifically, this process is formulated as:

$$Z_1 = \text{ReLU} \left(W_e \mathcal{X}_{\text{aux,sub}}^{(s)} \right), \quad (6)$$

$$Z_2 = \frac{1}{h} \sum_{t=1}^h Z_{1:t,:}, \quad (7)$$

$$Z_3 = \text{DGCN}(Z_2, \bar{A}_f^{(s)}, \bar{A}_b^{(s)}), \quad (8)$$

where $W_e \in \mathbb{R}^{(F-1) \times D}$ is the learnable encoding weight matrix. The resulting representations have the following dimensions: $Z_1 \in \mathbb{R}^{(n+u) \times h \times D}$, $Z_2 \in \mathbb{R}^{(n+u) \times D}$, and $Z_3 \in \mathbb{R}^{(n+u) \times D}$. This sequence transforms high-dimensional auxiliary features into compact, spatially-enriched node embeddings that capture both temporal patterns and structural dependencies.

Then, adaptive node weights are generated to modulate the importance of each node in the feature augmentation process. This is achieved by applying a learnable transformation followed by a sigmoid gating function to the enhanced node representations Z_3 . Specifically:

$$w_n = \sigma(W_a Z_3), \quad (9)$$

$$w_n^{\text{exp}} = w_n \otimes \mathbf{1}_{1 \times h}, \quad (10)$$

where $W_a \in \mathbb{R}^{D \times D}$ is a learnable weight matrix, σ denotes the sigmoid function, and $\mathbf{1}_{1 \times h} \in \mathbb{R}^h$ is a vector of ones. The resulting tensors have dimensions: $w_n \in \mathbb{R}^{(n+u) \times D}$ and $w_n^{\text{exp}} \in \mathbb{R}^{(n+u) \times h \times D}$, where the outer product $\otimes$ denotes dimension broadcasting—specifically, w_n is expanded along the temporal axis to replicate the node-specific weights across h time steps. This yields a time-aligned weight tensor w_n^{exp} that enables adaptive, node-wise feature scaling in the spatio-temporal domain.

The expanded node weights w_n^{exp} are then applied to the encoded auxiliary features Z_1 through element-wise multiplication, resulting in adaptively scaled representations:

$$Z_1^{\text{enh}} = Z_1 \odot w_n^{\text{exp}}, \quad (11)$$

where $Z_1 \in \mathbb{R}^{(n+u) \times h \times D}$, $w_n^{\text{exp}} \in \mathbb{R}^{(n+u) \times h \times D}$, and the enhanced output $Z_1^{\text{enh}} \in \mathbb{R}^{(n+u) \times h \times D}$. The Hadamard product $\odot$ performs channel-wise modulation, scaling the feature values at each node, time step, and dimension according to the learned adaptive weights. This operation preserves the original tensor structure while emphasizing or suppressing node representations based on their contextual importance derived from auxiliary information.

To generate a compact, interpretable augmentation signal, a projection head is applied to the enhanced feature tensor Z_1^{enh} . This head, parameterized by a learnable weight matrix $W_p \in \mathbb{R}^{D \times 1}$, reduces the feature dimension to a single scalar per node and time step:

$$\Delta^{(s)} = W_p Z_1^{\text{enh}}, \quad (12)$$

$$\tilde{Y}_{\text{sub}}^{(s)} = \hat{Y}_{\text{sub}}^{(s)} + \Delta^{(s)}, \quad (13)$$

where $Z_1^{\text{enh}} \in \mathbb{R}^{(n+u) \times h \times D}$, $\Delta^{(s)} \in \mathbb{R}^{(n+u) \times h}$, and the final enhanced reconstruction $\tilde{Y}_{\text{sub}}^{(s)} \in \mathbb{R}^{(n+u) \times h}$. The first operation computes a weighted sum across the feature dimension D for each node-time pair, effectively contracting the tensor to a scalar field $\Delta^{(s)}$ that encodes auxiliary-driven corrections. The second equation fuses this signal additively with the DGCN output $\hat{Y}_{\text{sub}}^{(s)}$, enabling continuous refinement of the reconstructed pH values.

This fusion strategy ensures that auxiliary information—such as environmental context or sensor metadata—is adaptively integrated into the imputation process. Each node's reconstruction is modulated by its own history of auxiliary features, allowing the model to adjust the imputation gain based on contextual relevance. This is particularly effective for unobserved nodes, where auxiliary signals provide critical inductive bias for accurate pH estimation.

2.6. Loss Function

The model is optimized using the mean absolute error (MAE) over the unobserved nodes, measuring the discrepancy between predictions and ground truth across all samples, time steps, and missing entries. The objective is formulated as:

$$\mathcal{L} = \frac{1}{S \cdot u \cdot h} \sum_{s=1}^S \sum_{i=1}^u \sum_{t=1}^h \left| Y_{\text{sub},i,t}^{(s)} - \tilde{Y}_{\text{sub},i,t}^{(s)} \right|, \quad (14)$$

where $Y_{\text{sub},i,t}^{(s)}$ denotes the true pH value at unobserved node i , time step t , and sample s , and $\tilde{Y}_{\text{sub},i,t}^{(s)}$ is the corresponding reconstructed output from the model.

258 **3. Experiments**

259 **3.1. Datasets**

260 This study uses a spatiotemporal water-quality dataset Zhao, Gkountouna and Pfoser (2019) with daily observations
261 from January 28, 2016, to January 1, 2018, at 37 monitoring stations in Georgia, USA. The dataset includes 12
262 variables: one pH imputation target and 11 auxiliary variables for modeling, listed as follows:

- 263 1. pH (Median): Median of field-measured pH values for unfiltered water samples, the target variable to be imputed.
- 264 2. pH (Max): Maximum pH, water, unfiltered, field, standard units.
- 265 3. pH (Min): Minimum pH, water, unfiltered, field, standard units.
- 266 4. SpCnd (Max): Maximum specific conductance, water, unfiltered, microsiemens per centimeter at 25 degrees
267 Celsius.
- 268 5. SpCnd (Min): Minimum specific conductance, water, unfiltered, microsiemens per centimeter at 25 degrees
269 Celsius.
- 270 6. SpCnd (Mean): Mean specific conductance, water, unfiltered, microsiemens per centimeter at 25 degrees Celsius.
- 271 7. DO (Max): Maximum dissolved oxygen, water, unfiltered, milligrams per liter.
- 272 8. DO (Min): Minimum dissolved oxygen, water, unfiltered, milligrams per liter.
- 273 9. DO (Mean): Mean dissolved oxygen, water, unfiltered, milligrams per liter.
- 274 10. Temp (Max): Maximum water temperature, degrees Celsius.
- 275 11. Temp (Min): Minimum water temperature, degrees Celsius.
- 276 12. Temp (Mean): Mean water temperature, degrees Celsius.

277 pH (Median) is the variable to be imputed, and the remaining 11 variables are used as auxiliary features. The
278 monitoring stations are located in two major watersheds—an inland watershed around Atlanta and a coastal watershed
279 in eastern Georgia—forming two spatially connected subnetworks with strong spatial dependence.

280 **3.2. Data Statistical Analysis**

281 This section provides a comprehensive analysis of the environmental data through visual and statistical methods.
282 The focus is on understanding temporal trends, spatial relationships, and variable interdependencies across multiple
283 sites.

284 The time series plots, shown in Figure 5, illustrate the trends of 12 variables, including the primary variable pH
285 (Median) and 11 auxiliary variables, for Site 37 training data over the period from 2016 to 2017. The pH-related
286 variables (pH (Median), pH (Max), pH (Min)) fluctuate between 0.66 and 0.95, showing high stability with notable
287 short-term peaks and troughs. SpCnd-related variables (SpCnd (Max), SpCnd (Min), SpCnd (Mean)) vary between
288 0.002 and 0.8, exhibiting strong fluctuations indicative of frequent external influences. DO-related variables (DO
289 (Max), DO (Mean), DO (Min)) range from 0.4 to 0.92, displaying a seasonal decline. Temperature-related variables
290 (Temp (Mean), Temp (Min), Temp (Max)) range from 0.2 to 0.8, showing clear seasonal patterns with higher values
291 in summer and lower in winter. Overall, all variables exhibit high-frequency small fluctuations overlaid with seasonal
292 variations, reflecting the dynamic nature of environmental data. Note that the time series plot is based on data from a
293 single site (Site 35) as a representative example for visualization.

294 The adjacency matrix heatmap, presented in Figure 6a, depicts the relationships among 37 sites, with the blank
295 diagonal indicating no self-connections and the deep blue regions signifying strong connections (edge weights is 1),
296 while lighter edges suggest weaker associations (edge weights is 0). This indicates a localized spatial connectivity,
297 with more frequent interactions among central sites.

298 The correlation heatmap, shown in Figure 6b, highlights the relationships of pH (Median) with other variables
299 across all sites in the training data. pH (Median) shows a positive correlation with pH (Max) (0.22) and pH (Min)
300 (0.15), indicating consistency across pH statistical measures. It exhibits a moderate positive correlation with SpCnd
301 (Max) (0.22) and a weaker positive correlation with SpCnd (Min) (0.20) and SpCnd (Mean) (0.29). The correlation
302 with DO variables (DO (Max) 0.86, may be useful for modeling. Last, a weak negative correlation with temperature
303 variables (Temp (Mean) -0.13, Temp (Min) -0.14, Temp (Max) -0.11), indicate minimal influence from temperature
304 changes. These correlations provide insights into the interdependencies centered around pH (Median).

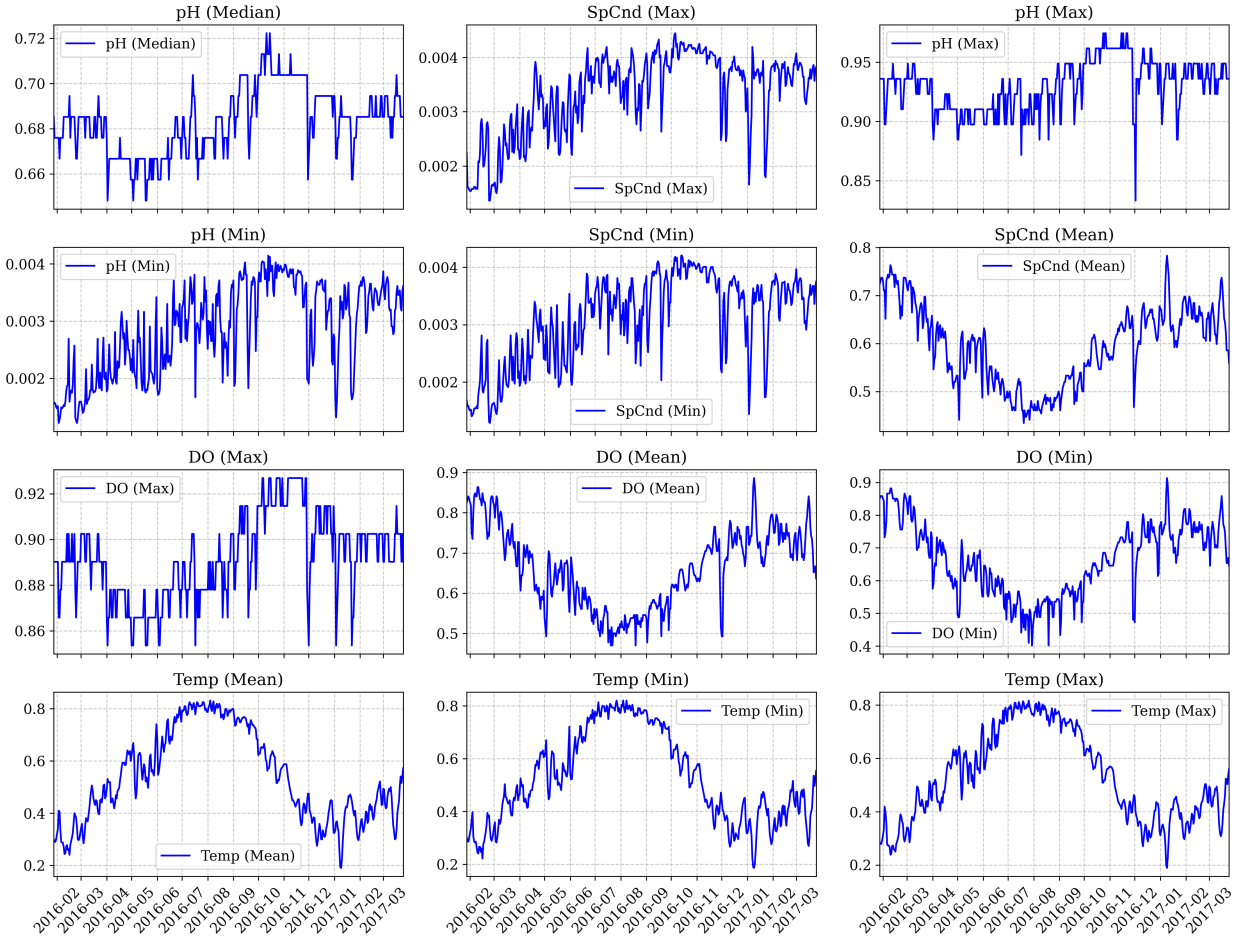


Figure 5: Time Series for Site 35 (Training Data) across 12 variables.

3.3. Evaluation Metrics

To objectively evaluate the performance of the pH value imputation model, this study employs three widely used regression error metrics: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Square Error (RMSE) (Hodson (2022); Chicco, Warrens and Jurman (2021); Willmott and Matsuura (2005)).

The Mean Absolute Error (MAE) measures the average magnitude of absolute errors between predicted and true values. It is relatively robust to outliers and is defined as:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (15)$$

where y_i denotes the true pH value of the i -th sample, $\hat{y}_i$ denotes the corresponding imputed value, and N is the total number of samples.

The Mean Absolute Percentage Error (MAPE) expresses the average error as a percentage, which facilitates comparison across different scales. It is calculated as:

$$MAPE = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (16)$$

Note that MAPE is undefined when any true value y_i is zero.

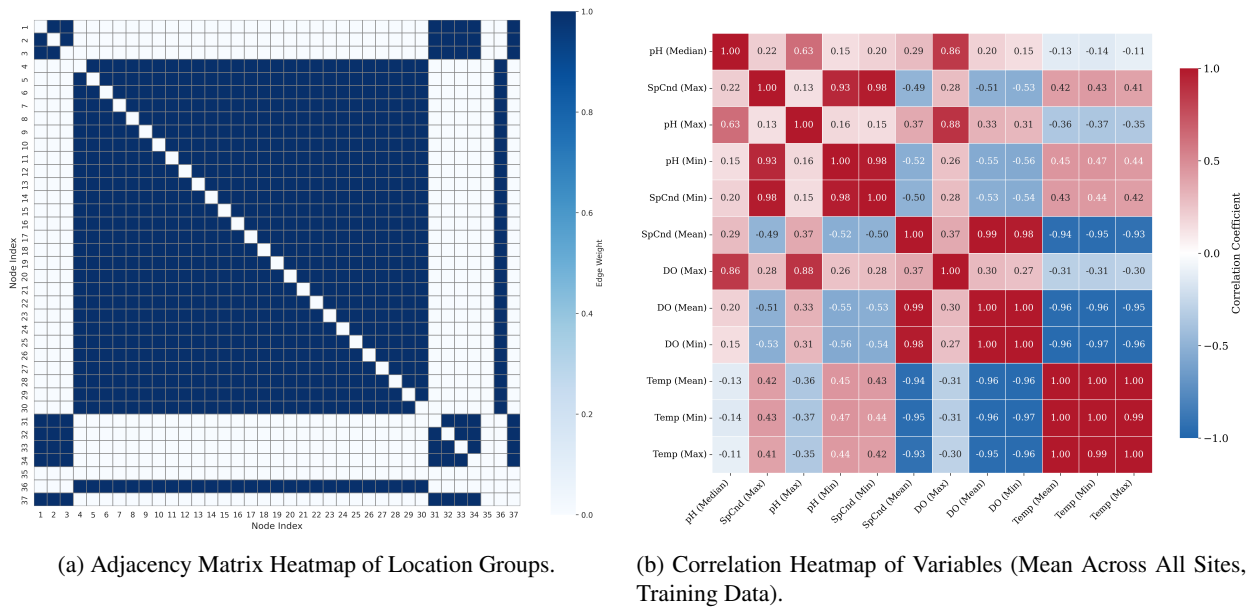


Figure 6: heatmaps

The Root Mean Square Error (RMSE) represents the square root of the average of squared errors. It penalizes larger errors more heavily and is therefore sensitive to outliers, making it a good indicator of prediction stability:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (17)$$

Lower values of MAE, MAPE, and RMSE indicate better imputation accuracy. These metrics are computed across all missing entries to comprehensively assess the model's performance.

3.4. Experimental Design

To comprehensively evaluate the proposed model for kriging imputation of water pH values, we design a series of experiments to verify its effectiveness, interpretability, and robustness. The experimental design includes four main components:

- Performance Comparison with Baselines:** We compare the proposed model against multiple baseline models from different methodological paradigms in terms of imputation accuracy and convergence speed. We further analyze the reasons behind performance differences, emphasizing how our model better captures spatial-temporal dependencies.
- Individual Contribution of Auxiliary Variables:** To evaluate the effectiveness of each auxiliary variable, we perform experiments where only one auxiliary variable is incorporated at a time. This allows us to quantify the individual contribution of each variable to the overall model performance.
- Ablation Study on Variable Combinations via Backward Selection:** We conduct a backward stepwise selection procedure to explore the optimal combination of auxiliary variables, providing insights into feature interactions and redundancy within the auxiliary set.
- Hyperparameter Sensitivity Analysis:** We analyze the sensitivity of the model to critical hyperparameters to assess its robustness and stability under varying configurations.

To ensure comprehensive and fair comparative experiments, a set of representative baseline models was selected to cover statistical methods, distance-based spatial interpolation, deep temporal modeling, dynamic graph neural networks, and hybrid spatio-temporal architectures. This design was intended to evaluate the effectiveness of the proposed approach under different modeling paradigms.

Table 2
Model Configurations

Parameter	Value
Time step length (h)	24
Subgraph nodes (N_{sub})	25
Unobserved nodes (u)	1

Table 3
Training Hyperparameters

Hyperparameter	Value
Batch size	128
Learning rate (lr)	0.001
Early stopping patience	20 epochs
Learning rate patience	20 epochs
Learning rate reduction factor	0.5

1. **Mean**: A simple temporal averaging strategy that is widely used as a lower-bound reference, reflecting the basic reconstruction performance when spatial-temporal dependencies are not modeled (Barnett, Van Der Pols and Dobson (2005)).
2. **KNN**: A spatial interpolation method based on K-nearest neighbors, which is capable of capturing local spatial correlations and is frequently applied to environmental monitoring with irregular sensor deployments (Guo, Wang, Bell, Bi and Greer (2003)).
3. **LSTM**: A Long Short-Term Memory network that is employed to model long-range temporal dependencies, representing a mainstream deep learning approach for time-series forecasting (Yu, Si, Hu and Zhang (2019)).
4. **DGCN**: A Dynamic Graph Convolutional Network that is designed to learn graph structures adaptively to capture evolving spatial dependencies, and is suitable for sensor networks with spatial heterogeneity (Zheng et al. (2021)).
5. **DGCN_LSTM**: A hybrid architecture in which DGCN and LSTM are combined to jointly model spatial and temporal dynamics, and is often adopted as a strong baseline for spatial-temporal prediction tasks (Ma, Fan, Yao and Zhang (2021)).

To ensure fair evaluation and avoid any label leakage, we follow the same temporal and nodal partitioning strategy as in the INCREASE Zheng, Fan, Wang, Qi, Chen and Chen (2023). Specifically, the entire 705 days of continuous observations are ordered chronologically and split into non-overlapping training, validation, and testing phases: the first 423 days constitute the training-validation period, within which the last 20% of days are held out as a validation set for hyperparameter tuning and early stopping, and the remaining 80% are used as the training set for parameter updates; the subsequent 282 days form an independent test set that is only used for final performance evaluation after training is completed. At the location level, 7 out of the 37 monitoring locations are designated as truly unobserved monitoring locations and are completely excluded from both training and validation; their pH time series are only used as targets in the test phase. The remaining 30 locations are treated as observed monitoring locations and are used to construct random spatio-temporal subgraphs during training and validation under a masked-reconstruction scheme. Because the 7 test-only unobserved monitoring locations do not participate in any stage of model optimization, this design effectively prevents test-label information from leaking into the training process. The other configurations applied to all models are summarized in Table 2, ensuring a fair and robust comparison. And all models are trained under the same optimization settings, as detailed in Table 3.

For the Mean model, the imputed value for a node at a specific time point is computed as the average of all known nodes' values at that time. The KNN model uses $K = 5$ nearest neighbors to estimate missing values, focusing on local spatial correlations. The LSTM model is configured with a hidden layer dimension of 64 and two stacked layers to capture temporal dependencies. The DGCN model employs a hidden dimension of 64 and an order of 2 to capture dynamic spatial dependencies. The DGCN_LSTM hybrid integrates DGCN and LSTM components, using DGCN outputs as LSTM inputs to jointly model spatial-temporal relationships, with consistent sub-component configurations.

Table 4Performance Comparison of Models for Water pH Imputation (Test Set, Mean $\pm$ Std)

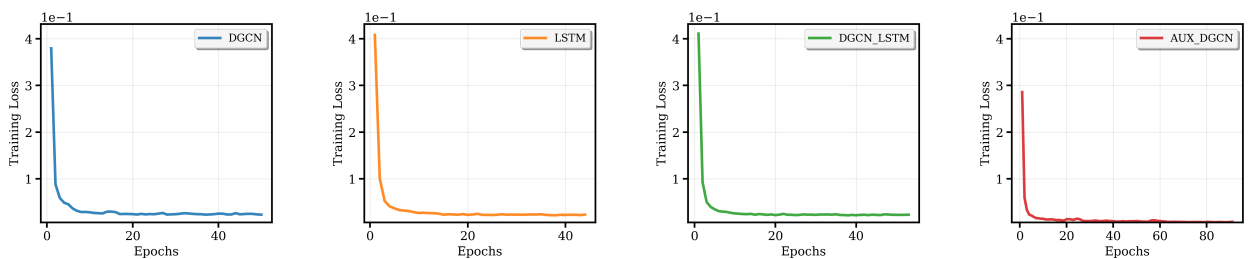
Model	MAE	MAPE (%)	RMSE
Mean	0.0176 $\pm$ 0.0000	2.6900 $\pm$ 0.0000	0.0227 $\pm$ 0.0000
KNN	0.0210 $\pm$ 0.0000	3.1900 $\pm$ 0.0000	0.0271 $\pm$ 0.0000
LSTM	0.0196 $\pm$ 0.0010	3.0000 $\pm$ 0.1600	0.0250 $\pm$ 0.0012
DGCN	0.0196 $\pm$ 0.0006	2.9800 $\pm$ 0.1000	0.0250 $\pm$ 0.0008
DGCN_LSTM	0.0192 $\pm$ 0.0005	2.9500 $\pm$ 0.0800	0.0246 $\pm$ 0.0007
Our Model	0.0084 $\pm$ 0.0008	1.2700 $\pm$ 0.1200	0.0119 $\pm$ 0.0009

4. Results

4.1. Performance Comparison with Baselines

In our experiments, to ensure robustness, we repeat each experiment five times with different random seeds and report the mean and standard deviation of all evaluation metrics. As shown in Table 4, our model achieves the best performance across all metrics, with an MAE of 0.0084, MAPE of 1.2700%, and RMSE of 0.0119. Compared to the second-best model, DGCN_LSTM, it reduces MAE, MAPE, and RMSE by 56.0%, 57.0%, and 51.6%, respectively. These results suggest that our model is effective for reconstructing missing pH values compared with the baselines.. The baseline models show varying performance: Mean and KNN exhibit higher errors (MAE of 0.0176 and 0.0210, respectively), indicating limited ability to capture dynamics. LSTM and DGCN perform similarly with an MAE of 0.0196, while DGCN_LSTM improves slightly to an MAE of 0.0192, suggesting incremental benefits from combining spatial and temporal features.

The convergence behavior of the models is illustrated in Figure 7. Our model demonstrates the fastest and most stable convergence, reaching a low training loss within approximately 20 epochs and maintaining it thereafter. DGCN and DGCN_LSTM also converge relatively quickly, stabilizing after around 20-30 epochs, though their final loss values remain higher than our model. LSTM shows a slightly slower convergence, stabilizing after 30 epochs, while its loss remains elevated compared to the graph-based models. This rapid and stable convergence of our model highlights its efficiency in optimizing the model parameters for pH imputation.

**Figure 7:** Training Loss Convergence for DGCN, LSTM, DGCN_LSTM, and our model over Epochs

To elucidate the factors contributing to our model's superior performance and convergence, we analyze the baseline models' limitations in capturing pH dynamics:

- Mean and KNN:** The Mean and KNN methods exhibit limited accuracy, failing to capture spatiotemporal dynamics. Notably, KNN performs worse than Mean in MAE (0.0210 vs. 0.0176), suggesting that spatial proximity alone is insufficient for pH imputation in heterogeneous watersheds.
- LSTM and DGCN:** LSTM captures temporal patterns but performs similarly to DGCN (MAE: 0.0196 for both) with moderate convergence, indicating that temporal modeling without explicit spatial context is inadequate.

398 3. **DGCN and DGCN_LSTM:** DGCN and DGCN_LSTM leverage spatial graph structures, outperforming purely
 399 temporal models with better convergence, but their lack of auxiliary physicochemical variables limits their
 400 representational power.

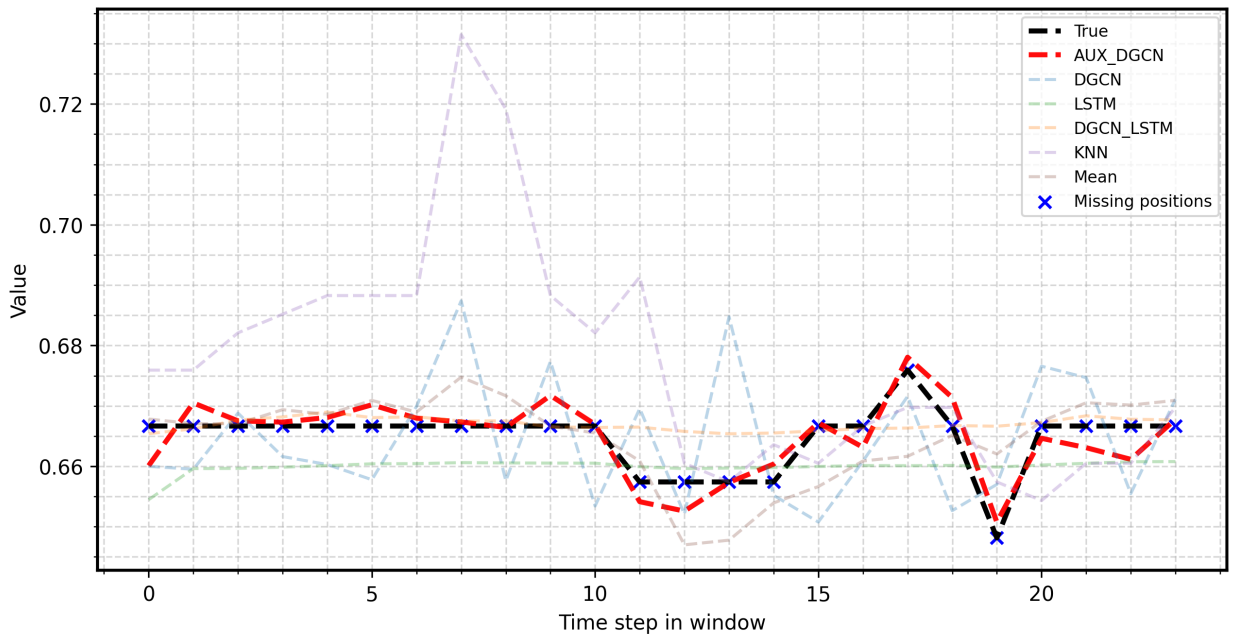


Figure 8: Time-series comparison between reconstructed and ground-truth normalized pH at two representative unmonitored stations in the test set. The left and right panels correspond to two different stations.

401 Furthermore, Figure 8 presents time-series comparisons between reconstructed and ground-truth normalized pH
 402 values at a representative unmonitored stations in the test set. In the case, our model (AUX_DGCN in the legend,
 403 hereafter referred to as our model) closely follows the ground-truth series and accurately captures both smooth
 404 trends and sharp local fluctuations. In contrast, the other baselines either exhibit noticeable bias (e.g., consistently
 405 overestimating or underestimating the level) or produce over-smoothed trajectories that fail to reproduce short-term
 406 variations. These examples qualitatively confirm that, rather than merely achieving good aggregate error statistics, our
 407 model can reliably reconstruct the temporal evolution of pH at locations without direct observations.

408 In summary, the results demonstrate that fusing auxiliary variables with dynamic spatial topology significantly
 409 improves pH imputation accuracy and convergence speed. our model provides a robust and interpretable framework
 410 for spatiotemporal environmental data reconstruction, with broad applicability in water resource management.

411 4.2. Individual Contribution of Auxiliary Variables

412 To systematically identify the most effective predictors for pH imputation, we assess the standalone performance
 413 of 11 physicochemical water quality variables. Each variable is evaluated as the sole auxiliary input within a consistent
 414 model architecture, isolating its individual contribution.

415 As depicted in Figure 9, DO(Max) emerges as the most informative variable, with an MAE of 0.0081, MAPE of
 416 1.2225%, and RMSE of 0.0118, significantly outperforming other variables. This superior performance is consistent
 417 with our earlier correlation analysis, reflecting the strong coupling between peak DO and pH, may driven by
 418 photosynthetic CO₂ uptake during diurnal maxima, which concurrently increases both parameters. The low standard
 419 deviation in MAE (0.0004) underscores the high stability and reliability of DO(Max) as a proxy for pH fluctuations.

420 Other variables exhibit moderate performance: pH(Max) records an MAE of 0.0114 and MAPE of 1.7128%, while
 421 SpCnd(Max) shows an MAE of 0.0179 and MAPE of 2.7398%. These results indicate that historical pH values and
 422 ionic content offer useful predictive signals, though they are less direct than DO. Temperature metrics, particularly
 423 Temp(Max) with an MAE of 0.0201 and MAPE of 3.0859%, perform least effectively, likely due to their indirect

influence through thermal effects on chemical equilibria. This systematic evaluation highlights DO(Max) as the optimal single predictor, supported by both statistical metrics and biogeochemical insights.

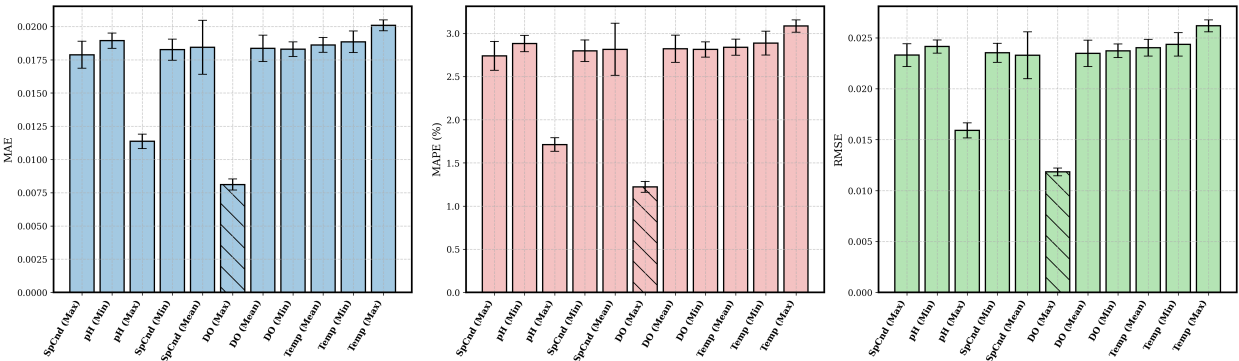


Figure 9: Model Performance Comparison for Individual Auxiliary Variables.

4.3. Ablation Study on Variable Combinations via Backward Selection

Exploring all possible combinations of 11 physicochemical auxiliary variables ($2^{11} = 2048$ subsets) for imputation is computationally infeasible. We thus employ backward elimination, initiating with the full set of variables and iteratively removing the least impactful variable based on performance degradation. This method evaluates whether a reduced subset can match the full model’s accuracy while elucidating the collective role of variables in our model.

Figure 10 presents the performance for the first iteration of backward elimination, where each variable is individually excluded from the full set. The complete 11-variable configuration achieves the lowest errors (MAE: 0.0070, MAPE: 1.0587%, RMSE: 0.0103). Removing any variable increases errors, indicating that all variables contribute meaningfully to the model’s performance. Consequently, we ceased further search for an optimal subset, as the degradation in performance upon removing any variable underscores the utility of the full set.

Additionally, removing DO (Max) results in the largest performance degradation, highlighting its significant complementary contribution. In contrast, excluding DO (Min) leads to the smallest performance drop, suggesting its minimal unique impact. These findings highlight that our model’s superior performance arises from integrating all auxiliary variables, leveraging their collective biogeochemical signals—such as pH-DO interactions during photosynthetic peaks.

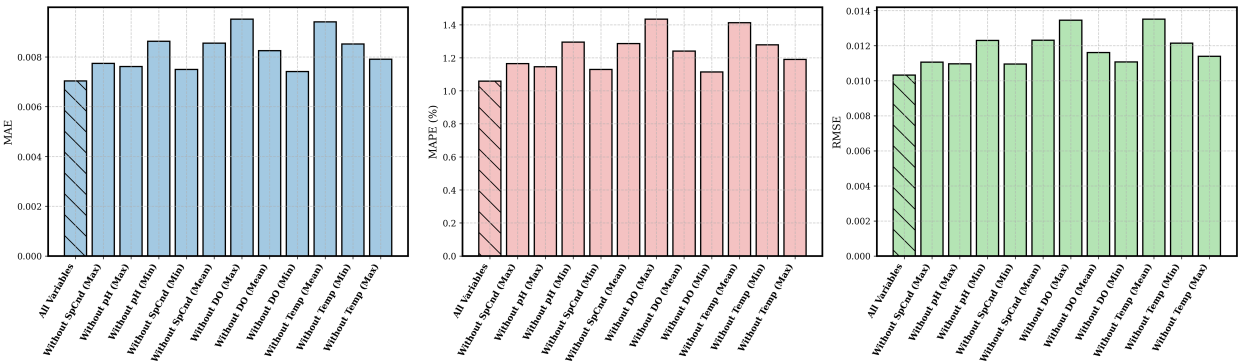


Figure 10: Performance Comparison of Auxiliary Variable Combinations from Backward Feature Selection.

4.4. Hyperparameter Sensitivity Analysis

To systematically examine the reliability of the proposed model for pH imputation under different data sampling and monitoring-network configurations, we conduct a sensitivity analysis on four key hyperparameters. Specifically,

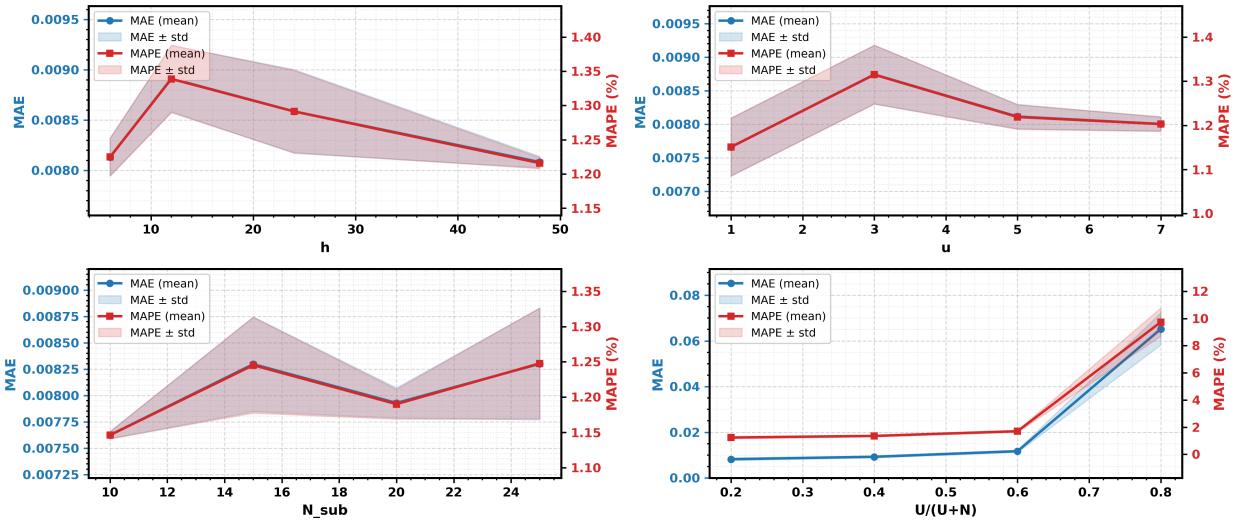
Table 5

Hyperparameter configurations for the sensitivity analysis experiments.

Tuned hyperparameter	Value range	Other hyperparameters
h	6, 12, 24, 48	$u = 1, N_{\text{sub}} = 25, U/(U + N) = 0.2$
u	1, 3, 5, 7	$h = 12, N_{\text{sub}} = 25, U/(U + N) = 0.2$
N_{sub}	10, 15, 20, 25	$h = 12, u = 1, U/(U + N) = 0.2$
$U/(U + N)$	0.2, 0.4, 0.6, 0.8	$h = 12, u = 1, N_{\text{sub}} = 25, 18, 12, 6^\dagger$

[†] For the experiments that vary the global proportion of unmonitored stations $U/(U + N)$, the subgraph size N_{sub} is also adjusted from 25 to 6 so that the ratio N_{sub}/N remains approximately consistent as the monitoring network becomes sparser, given that the total number of stations is $N = 37$.

we vary the forecasting horizon h , the number of unobserved nodes u in each sampled subgraph, the subgraph size N_{sub} , and the global proportion of unmonitored stations $U/(U + N)$. These hyperparameters respectively control the temporal window length, the sparsity of observations within a subgraph, the spatial coverage of each subgraph, and the overall sparsity of the monitoring network. Through this sensitivity analysis, we evaluate the stability of imputation accuracy under different monitoring configurations and missing-data levels, thereby assessing the overall robustness of the model. The hyperparameter configurations used in the sensitivity analysis experiments are summarized in Table 5, where only one hyperparameter is varied at a time while the others are kept fixed.

**Figure 11:** Performance across different hyperparameter settings.

For the sensitivity experiments with respect to the forecasting horizon h , the number of unobserved nodes u in each sampled subgraph, and the subgraph size N_{sub} , the corresponding error curves remain relatively flat, as shown in Figure 11. The MAE and MAPE values and their uncertainty bands remain close across different settings, and the apparent oscillations are mainly an artifact of the narrow vertical scale rather than genuine performance variation. These observations indicate that the model is relatively insensitive to the choice of temporal window length, the sparsity of observations within a subgraph, and the local spatial coverage scale, and can maintain stable imputation accuracy over a wide range of hyperparameter values.

For the experiments that vary the global proportion of unobserved monitoring locations $U/(U + N)$, the results in Figure 11 reveal a more pronounced trend. As $U/(U + N)$ increases from 0.2 to 0.6, the errors grow only gradually and remain at a comparable magnitude to other configurations. When the proportion is further increased to 0.8, both the mean errors and their uncertainty exhibit a marked escalation, suggesting that under such extremely sparse monitoring conditions the network approaches the upper sparsity limit at which the model can still support high-quality imputation.

463 Taken together, these findings demonstrate that the proposed model exhibits strong robustness under diverse
464 configurations of temporal windows, observation sparsity patterns, spatial subgraph scales, and monitoring-network
465 sparsity, with only expected performance degradation arising under extreme sensor sparsity.

466 **4.5. Discussion**

467 The proposed water quality auxiliary-enhanced spatio-temporal imputation framework based on diffusion graph
468 convolutional networks not only achieves state-of-the-art accuracy in reconstructing missing pH data but also provides a
469 novel paradigm for integrating heterogeneous environmental signals into graph-based learning frameworks. Its superior
470 performance demonstrates that the fusion of diffusion graph convolution with node-level auxiliary enhancement can
471 effectively capture both global spatial dependencies and local physicochemical dynamics, overcoming the limitations
472 of conventional Kriging or tensor completion methods.

473 From a methodological perspective, our model establishes a scalable and inductive framework that can dynamically
474 adapt to evolving monitoring networks. By employing subgraph sampling, the model remains computationally efficient
475 even when sensor networks expand or experience failures, making it particularly suitable for long-term environmental
476 monitoring systems. Furthermore, the auxiliary variable enhancement mechanism introduces interpretability by
477 revealing which indicators (e.g., dissolved oxygen and conductivity) most strongly influence imputation outcomes,
478 thus offering valuable insight into the underlying biogeochemical processes governing water quality dynamics.

479 In terms of practical implications, this approach provides a solid foundation for constructing real-time intelligent
480 monitoring and early-warning systems for aquatic environments. The ability to accurately restore missing water quality
481 data can significantly improve downstream tasks such as pollution source identification, risk forecasting, and ecological
482 health assessment. With proper integration into Internet of Things (IoT) platforms and edge computing architectures,
483 our model could support adaptive monitoring networks capable of autonomous decision-making and self-correction
484 under uncertain or incomplete data conditions.

485 **5. Conclusions**

486 This study has introduced the Auxiliary-Enhanced Spatio-Temporal Imputation Network utilizing Diffusion Graph
487 Convolutional Networks to address the challenge of missing water quality data. By integrating auxiliary variables such
488 as dissolved oxygen, turbidity, and conductivity, the proposed model effectively captures complex spatio-temporal
489 dependencies. The main conclusions drawn from this study are as follows.

- 490 1. **Model innovation and framework advantages.** This study proposes an auxiliary-enhanced spatio-temporal
491 imputation framework based on Diffusion Graph Convolutional Networks, integrating subgraph sampling
492 and node-level adaptive augmentation. This design enables inductive generalization to dynamic graph while
493 efficiently fusing multi-source physicochemical indicators.
- 494 2. **Superior performance and robustness across experiments.** In experiments, the proposed model achieves the
495 best performance among all baselines, reducing MAE from 0.0192 (best baseline, DGCN_LSTM) to **0.0084**,
496 and MAPE from 2.95% to **1.27%**. It also maintains stable accuracy under varying forecasting horizons, subgraph
497 sizes, the number of unobserved nodes in sampled subgraph and global proportion of unobserved monitoring
498 locations, demonstrating strong robustness.
- 499 3. **Significant contribution of auxiliary variables.** The auxiliary-variable analysis reveals that **DO(Max)** con-
500 tributes the most to pH reconstruction, achieving an MAE of 0.0081 when used alone. The optimal auxiliary-
501 variable configuration is the **full 11-variable set**, as removing any variable leads to performance degradation,
502 confirming the complementary effects among indicators.
- 503 4. **Practical implications and extensibility.** Owing to its inductive capability and real-time adaptability, the
504 proposed framework can be seamlessly integrated into large-scale IoT-based aquatic monitoring systems for
505 tasks such as automated anomaly detection, early-warning, and digital-twin water environment management. It
506 also provides a generalizable paradigm for other environmental variables such as turbidity, conductivity, or air
507 quality metrics.

508 The successful application of auxiliary variables to generate node-level enhancement signals marks a significant
509 advancement in water quality monitoring, offering a scalable and efficient solution for data imputation. Future research
510 could explore the integration of real-time sensor data and the extension of this framework to other environmental
511 parameters, further improving its applicability in dynamic aquatic ecosystems.

512 **CRedit authorship contribution statement**

513 **Jiayu Fan:** Methodology, Writing, Experiments, editing. **Lewei Xie:** Conceptualization, Methodology, Ex-
514 periments. **Mengnan Shi:** Conceptualization, Writing, review. **Qiang Yao & Hongtao Li:** Funding acquisition,
515 Investigation, Supervision, Resources.

516 **Declaration of Competing Interest**

517 The authors declare that they have no known competing financial interests or personal relationships that could have
518 appeared to influence the work reported in this paper.

519 **Declaration of Generative AI and AI-Assisted Technologies in the Writing Process**

520 During the preparation of this work, the authors used ChatGPT and DeepL in order to polish the language of the
521 manuscript. After using these tools, the authors reviewed and edited the content as needed and take full responsibility
522 for the content of the publication.

523 **Acknowledgment**

524 This work is supported by the Postdoctoral Fellowship Program of CPSF (GZB20240503) and the China
525 Postdoctoral Science Foundation(2025M773161)

526 **Data Availability**

527 The datasets and source code used in this study are fully open and available for use. For details, please refer to the
528 following link: <https://github.com/Xielewei/Water-Quality.git>

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